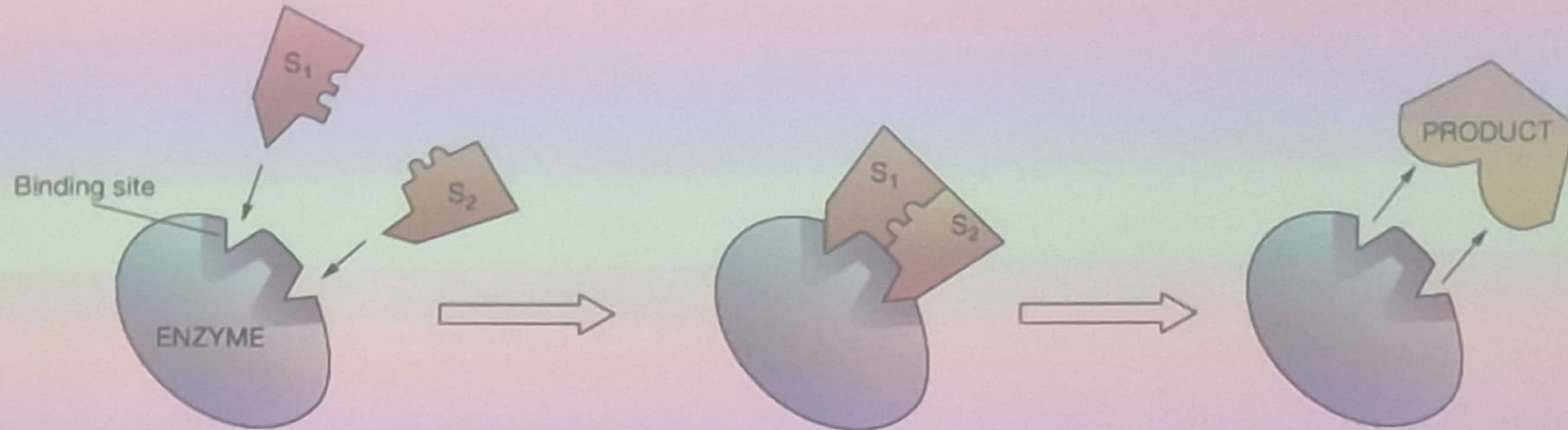


Specificity: a) lock and key or b) induced fit.

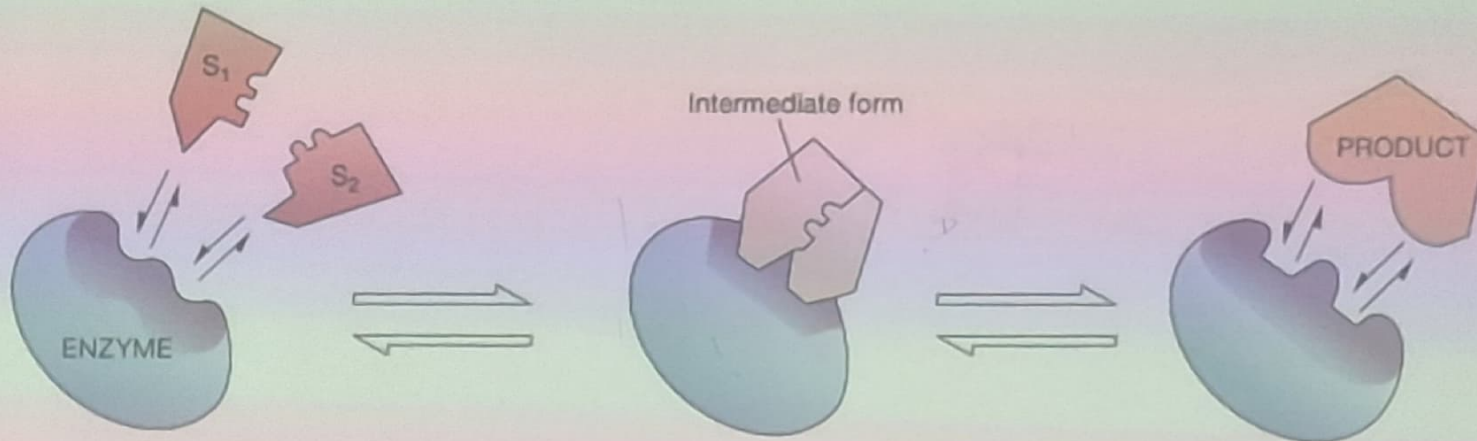
(a) Lock-and-key model

In this older model, the enzyme binding site was an exact match for the substrates.



(b) Induced-fit model

In this model, the enzyme binds most tightly to an intermediate form of substrate or product. This model is more compatible with reversible reactions that are catalyzed by one enzyme.



Classification and Nomenclature

1. Often named by adding the suffix *-ase to the name of* the substrate upon which they acted.
2. Most enzymes end with ase and typically their name indicates their function or the substrates they bind.



2. Most enzymes end with ase and typically their name indicates their function or the substrates they bind.

e.g. Urease, DNA Polymerase

Naming of Enzymes

Some important examples:

kinase - adds phosphates to molecules;

phosphotase - removes phosphates from molecules;

dehydrogenase - removes Hydrogens;

hydrolase - adds H_2O ;

synthase - synthesis reactions;

carbonic anhydrase - removes H_2O from carbonic acid;

amylase - digests starch;

lipase - digests lipids;

protease - digests proteins.

IUBMB nomenclature

1956 - to create a systematic basis for enzyme nomenclature

Four- digit numbered code

- first digit - major class
- Second digit - sub class
- third digit - sub sub class
- final digit - specific enzyme

Enzyme classification

Six classes

1. Oxidoreductase

Catalyze redox reactions, such as the transfer of electrons, hydrogen, or oxygen atoms from one substrate to another

e.g. Alcohol Dehydrogenase

Lactate dehydrogenase

Cytochrome oxidase

Note: Each class of enzyme catalyzes many different specific reactions, but they all carry out a general type of reaction.

2. Transferase

Transfer functional groups, like methyl or glycosyl groups, from one molecule to another

e.g. Kinases

Transaminase

Phosphorylase

Note: Each class of enzyme catalyzes many different specific reactions, but they all carry out a general type of reaction.

3. Hydrolase

Catalyze hydrolytic reactions, such as the hydrolysis of bonds in proteins, esters, and more

cleave C-O, C-N, C-S or P-O etc. bonds by adding water across the bond

e.g. Lipase

Acid Phosphatase

(important in digestive process)

Note: Each class of enzyme catalyzes many different specific reactions, but they all carry out a general type of reaction.

4. Lyases

Catalyze the removal or addition of functional groups from double-bonded carbons

cleave C-O, C-N, or C-S bonds but do so without addition of water and without oxidizing or reducing the substrates

e.g. Aldolase

Fumarase

Carbonic anhydrase

Note: Each class of enzyme catalyzes many different specific reactions, but they all carry out a general type of reaction.

5. Isomerase

catalyze intramolecular rearrangements of functional groups that reversibly interconvert to optical or geometric isomers

e.g. Triose isomerase

Phosphohexose isomerase

Mutase

Note: Each class of enzyme catalyzes many different specific reactions, but they all carry out a general type of reaction.

6. Ligase

Catalyze biosynthetic reactions that form a covalent bond between two substrates utilizing ATP-ADP interconversion

e.g. Glutamine synthetase
DNA- ligase

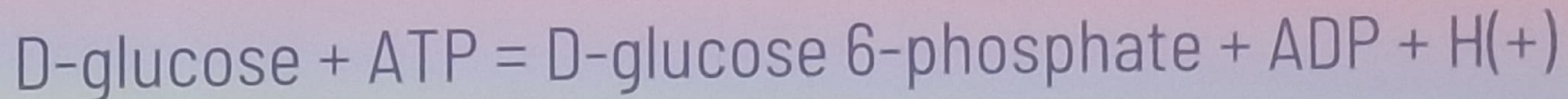
Note: Each class of enzyme catalyzes many different specific reactions, but they all carry out a general type of reaction.

2.7.1.2 : Glucokinase

2- class name (transferase)

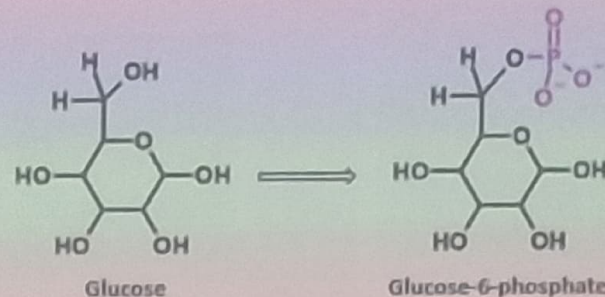
7- subclass name (transferring phosphorus-containing groups)

1- sub sub class (Phosphotransferases with an alcohol group as acceptor)



Enzyme Specificity - ability of enzyme to catalyze only certain reactions. Some are very specific, such as **glucokinase**.

It catalyzes this reaction:



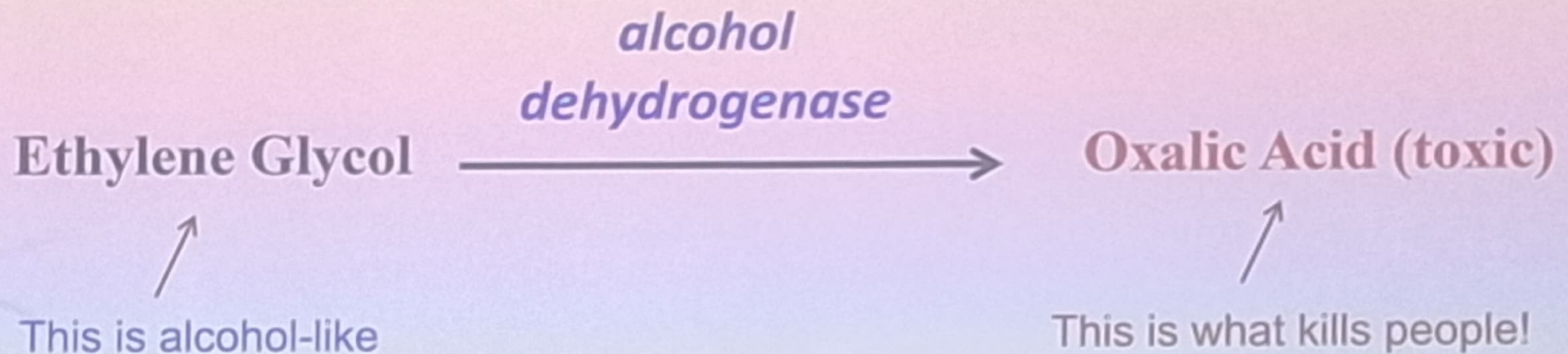
The type and amount of enzymes in the body can vary.

e.g., **alcohol dehydrogenase** removes hydrogens from ethanol ('alcohol') thereby reducing its toxic effects.

Q: Is everyone's ability to "handle" alcohol the same?

An interesting example of Competitive Inhibitors:

Ethylene glycol (antifreeze) is a poison which can kill people if ingested! This is what happens in the body:



Enzyme Activation

Some enzymes exists in an ***inactive*** state, need to be ***activated***.

1) Proteolytic Activation:

e.g. $\text{Pepsinogen} + \text{HCl} \longrightarrow \text{Pepsin}$
(inactive) (active)
(longer) (shorter)

Other Examples of Proteolytic Activation:

Fibrinogen - > becomes...

Thrombinogen -> becomes...

Angiotensinogen -> becomes ...

Enzyme Activation

2) Co-factors: Components required for substrate binding at the *active site*.

e.g., Zn^{2+} Ca^{2+} , Mg^{2+} or Cu^{2+} (conformational changes).

Cofactor	Example Enzymes
Zn^{2+}	Carbonic Anhydrase
Zn^{2+}	Alcohol Dehydrogenase
Cu^{2+}	Cytochrome Oxidase
K^{+} and Mg^{2+}	Pyruvate Phosphokinase

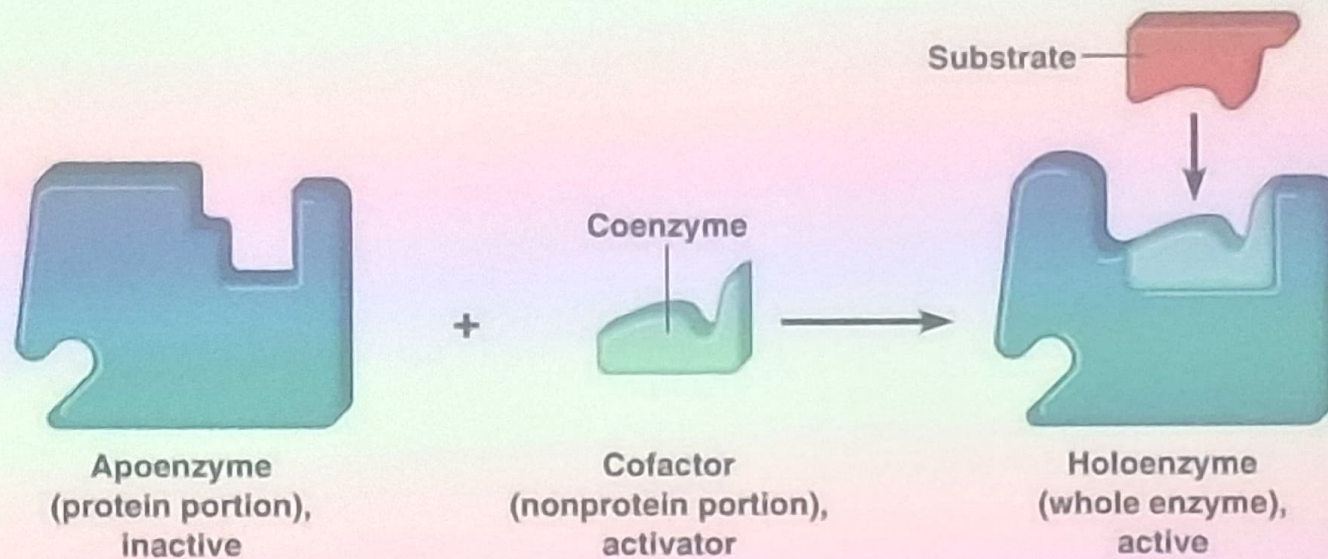
Cofactors provide additional reactive groups to the enzymes and thereby aid in catalysis.

A **cofactor** can be linked to the protein portion of the enzyme either covalently or non-covalently.

Some cofactors are simple metal ions and other cofactors are complex organic groups, which are also called **coenzymes**. Cofactors which are tightly associated with the protein covalently or non-covalently are called **prosthetic group**.

3) **Co-enzymes:** Small organic molecules to accept and transfer electrons (e^- s) from different enzymatic reactions.

Bind loosely to an enzyme at active site to help catalyze reactions.
Usually vitamins, vitamin derivatives, or from nucleotides.



Apoenzyme - an inactive enzyme that requires cofactor/coenzyme.

Holoenzyme - a complete and catalytically active enzyme because it has its cofactor/coenzyme.

e.g., Co-enzyme NAD^+ shuttles e^- s in glycolysis

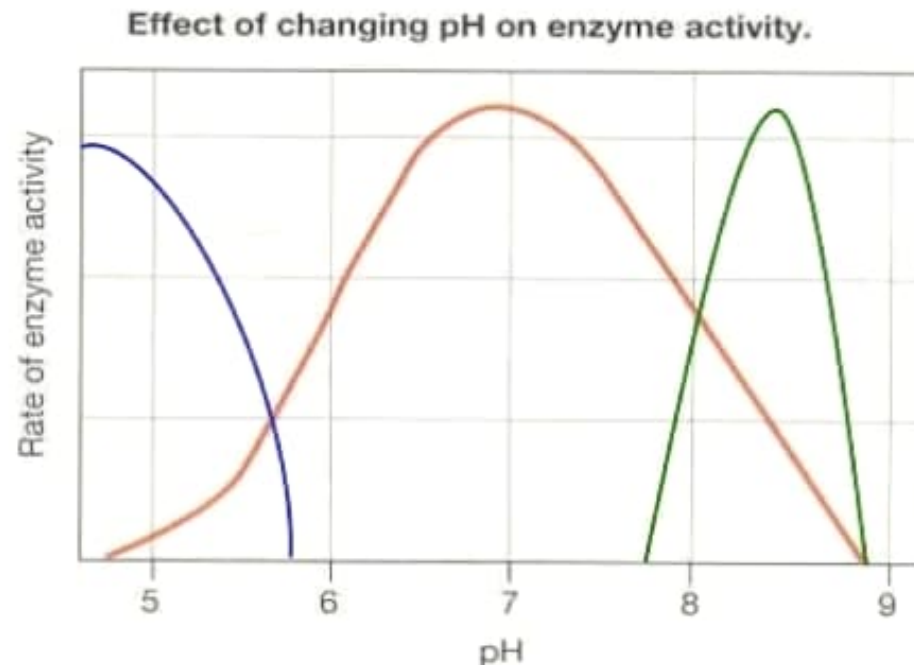
Factors that Effect Enzyme Activity

Once an enzyme is active, several factors can modulate (change) their activity.

1. pH (Acidity/Alkalinity)

Enzymes function within certain pH ranges. Changes in pH alter 3^o structure. Beyond a critical level (outside its optimal pH range), the enzyme is **denatured**.

e.g., compare enzymes of mouth, stomach and small intestine.



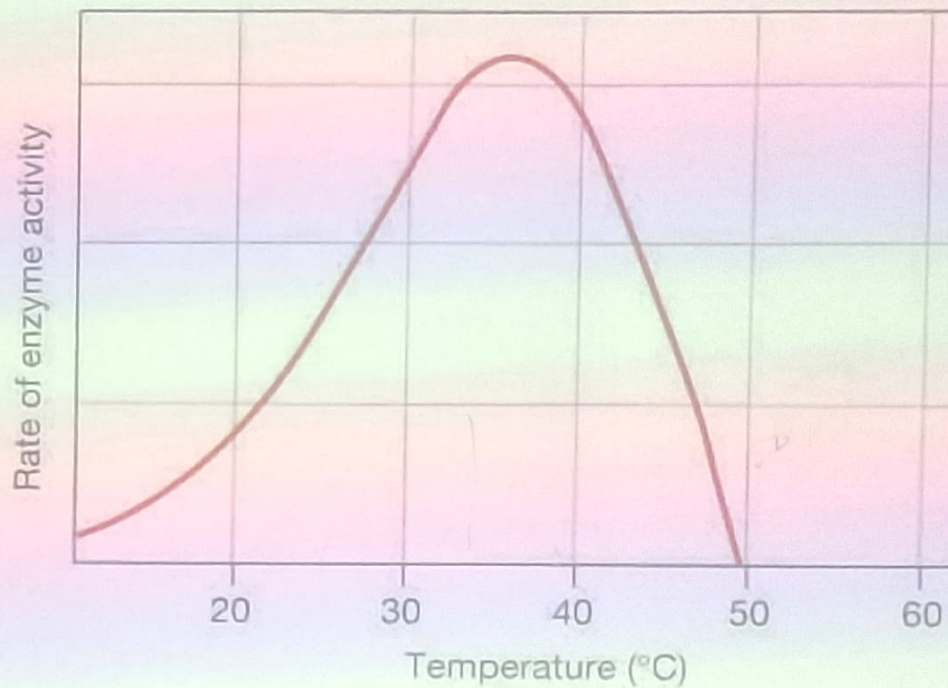
*Salivary amylase breaks down starch to maltose in the mouth, has an optimum pH 6.7 to 7.0.

2. Temperature

Enzymes function within certain and temperature ranges. Most human enzymes are optimal at 36-37°C. Again, beyond a critical level, enzymes are **denatured**.

e.g., think of the effects of a fever on enzymes in the body.

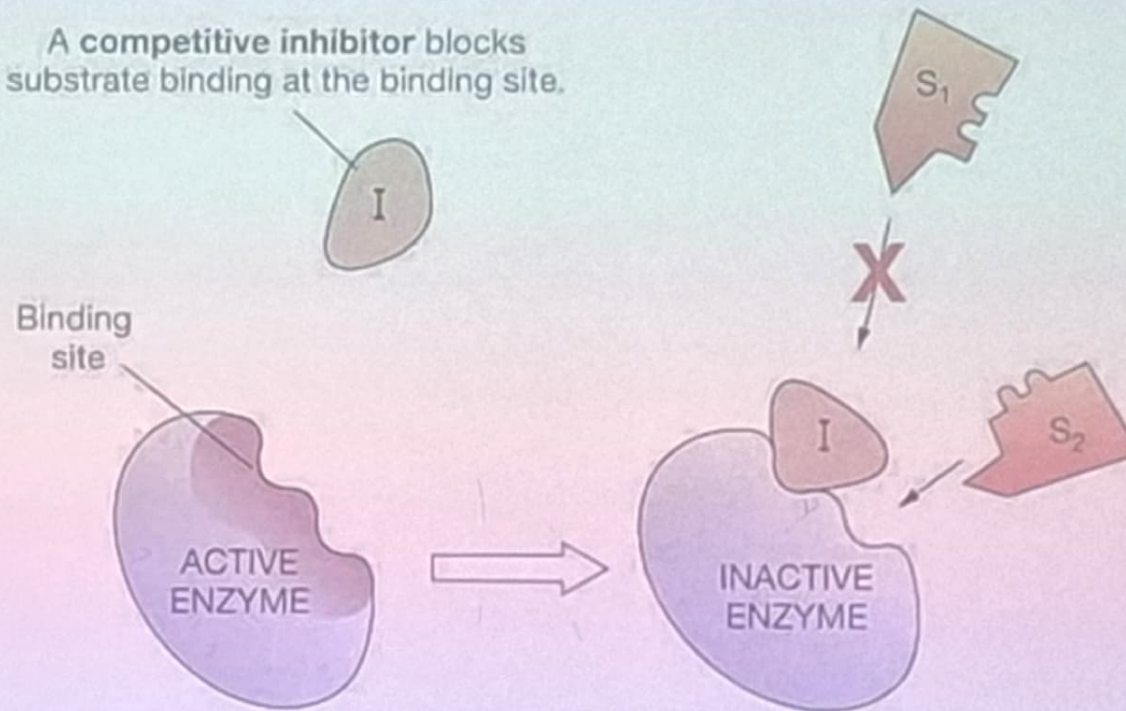
Effect of changing temperature on enzyme activity.



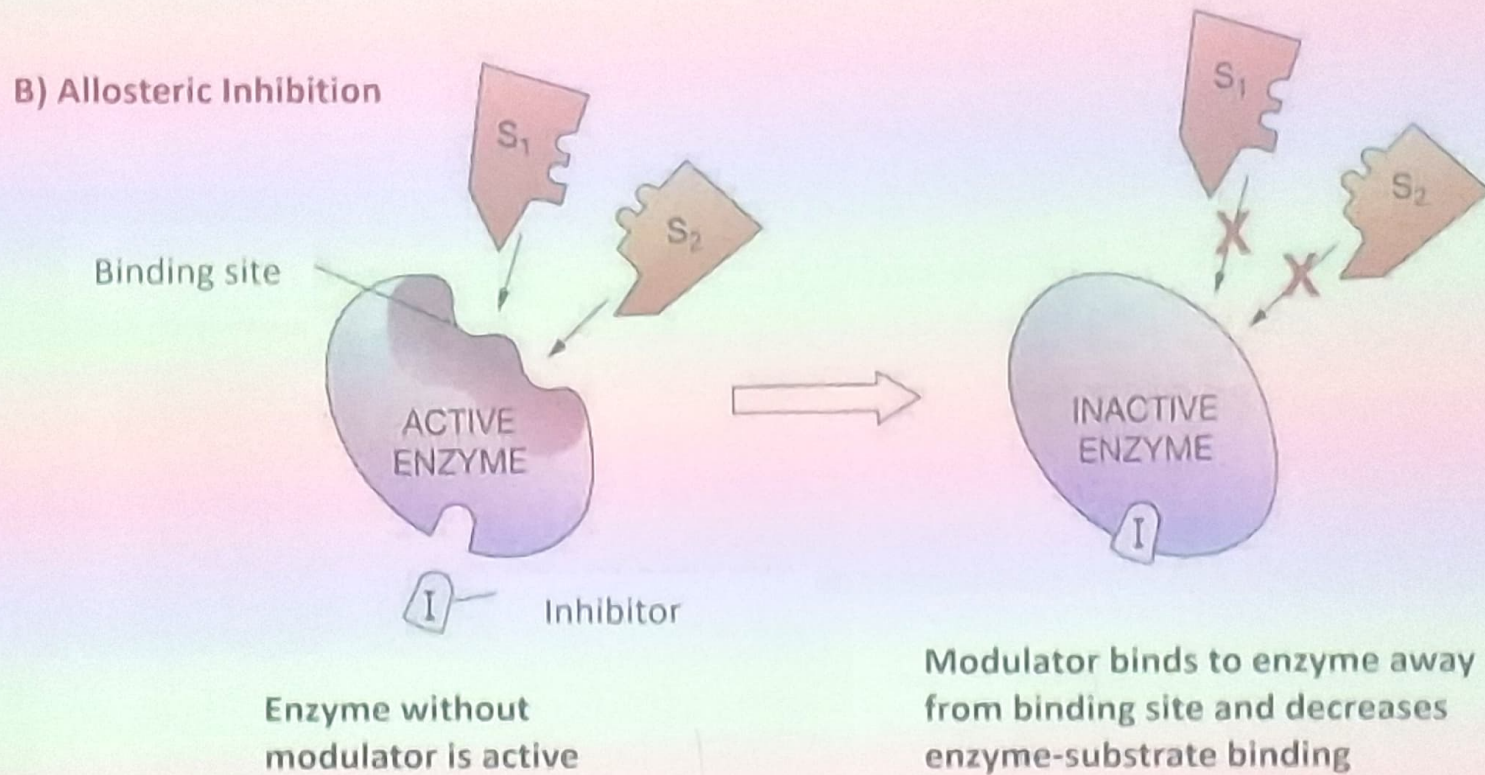
3. Inhibitors - Molecules that bind to enzymes and inhibit true substrate from binding

A) Competitive Inhibitors bind to the *active site* without being acted on, thus reducing reaction rate of true substrate(s). In other cases, the competing molecule is acted on by the enzyme, but again, inhibits reaction with natural substrate.

A competitive inhibitor blocks substrate binding at the binding site.



B) Non-Competitive Inhibitors bind *at some site other than the active site*. They do not affect enzyme-substrate binding but inhibit the enzyme from catalyzing the reaction. Some act by binding to the cofactors of enzymes.



Enzymes are Proteins that act as Biological Catalyst

They **speed up** rate of chemical reactions w/out being consumed in process.

The **active site** on the enzyme is where the substrates bind.

Substrates are the *reactants* in a chemical reaction requiring enzymes.

